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University of Rajshahi
(For Affiliated College)
Department of Computer Science and Engineering

B. Sc. (Engg.) Part-I Even Semester Examination 2021
Course: CSE 1221 (Object Oriented Programming)
Session:2020-2021
Full Marks: 60 Duration: 3 (Three) Hours

Answer 06 (Six) questions taking any 03 (Three) questions from each section

Section-A

1. a) Compare between Java and C++ programming languages. 3
   b) The following code gives the nth term of a Fibonacci series. Solve this problem with Java recursion. 2

```
int Fibonacci (int n) {
     int firstTerm=0, secTerm=1;
     int nextTerm;
     for (int i=1; i<=n; i++) {
           nextTerm=firstTerm+ secTerm;
           firstTerm= secTerm; secTerm= nextTerm;}
     return nextTerm;
}
```

   c) Why encapsulation is useful in Java Programming? 2
   d) What is meant by pass by reference and pass by value in Java? 3

2. a) What is an object in Java? Describe the characteristics of an object. 3
   b) Discuss the advantages and disadvantages of arrays. 2
   c) What is a constructor in Java? Explain the rules for creating a constructor. 3
   d) What if you write "static public void" instead of "public static void"? 1
   e) What is the output of the following code? 1

```
Public static void main (string [] args) {
      System.out.println (10+20+ "RUCSE");
      System.out.println ("RUCSE"+ 10+20);
}
```

3. a) How do you declare, instantiate and initialize multi-dimensional arrays? 2.5
   b) What are the default initialization values of elements in an array? 1.5
   c) Write a Java program to store 10 integers in an array. Again, ask user to give a number and tell user whether that number is present in array or not. 4
   d) What is the difference between method overloading and method overriding? 2

4. a) How access and non-access modifiers can be added to a class or class member declaration? 2.5
   b) What are the scopes of primitive variables. How long primitive variables exist in memory? 3.5
   c) How does Java programming pass primitive variables to methods-- by value or by reference? 2
   d) What are wrapper classes in Java programming language? 2

Section-B

5. a) Write a Java program to read 10 integers from an input file, then find the maximum value among the 10 integers, and write it to an output file. 3
   b) Describe bite streams, character streams, and standard streams with example. 3
   c) Write a Java program that creates InputStreamReader to read standard input stream until the user types a "q". 2
   d) Why would you need a statement that involves two different stream classes, PrintWriter and FileWriter? 2

6. a) Write a Java program that demonstrates exception handling in Java programming. 3
   b) Describe with an example how to handle multiple catch blocks in Java programming. 3
   c) What happens if you try to access an element with an index greater than the size of an array? 1
   d) What are final classes and final methods of Java programming language? 3

7. a) What are the differences between Swing and AWT? 2
   b) Distinguish between `Invokeandwait()` and `Invokelater()`. 2
   c) Write an AWT GUI application called AWTAccumulator, which has four components: 6
      i. a `java.awt.Label "Enter an integer and press enter";`
      ii. an input `java.awt.TextField;`
      iii. a `java.awt.Label "The accumulated sum is"`, and
      iv. a protected (read-only) `java.awt.TextField` for displaying the accumulated sum.

8. a) Describe different types of design patterns in Java. 3
   b) What are the advantages of Java design patterns? 2
   c) Briefly describe the factory design pattern. 2
   d) Describe bridge design pattern with example. 3

University of Rajshahi
(For Affiliated College)
Department of Computer Science and Engineering
B. Sc. Engg. Part-1 Even Semester Examination 2021
Course: CSE 1221 (Object Oriented Programming)
(Session: 2019-2020)
Full Marks: 52.5 Duration: 3 (Three) Hours
Answer 06 (Six) questions taking any 03 (Three) questions from each section

Section-A

1. a) What is object oriented programming (OOP)? Write the basic features of OOP. 3
   b) Describe encapsulation, inheritance, and polymorphism briefly. 3
   c) When do we declare a member of a class static? 2.75

2. a) Why are command line arguments passed as a string? 2.5
   b) Why doesn't the main method throw an error with no arguments? 2.25
   c) How can the static main method use instance variables? 2
   d) What are final classes and final methods in Java programming language? 2

3. a) What are the purposes of constructor and destructor function? Can these functions have input parameters and return type? 2.75
   b) Define inline function. Explain the different ways to declare a function as inline. 3
   c) What is the difference between method overloading and method overriding? 3

4. a) How access modifiers can be added to members of a class? 2.5
   b) Describe the object class in Java programming? 2.5
   c) Is it possible to have multiple main methods in the same class? 2
   d) What are wrapper classes in Java programming language? 1.75

Section-B

5. a) Describe Array and ArrayList with appropriate examples. 3
   b) Is it possible to have multiple catch blocks in Java programming? 2.75
   c) Can we keep other statements in between try, catch and finally blocks? Explain with example. 3

6. a) Write a Java program that reads 10 integers from an input file, find the smallest value among the 10 integers, and write it to an output file. 3
   b) Discuss the basic structure of exception handling in Java. 3
   c) How new and delete are different from malloc () and free ()? 2.75

7. a) What is thread? Discuss Java thread model. 3.25
   b) Write a program to demonstrate the creation and execution of threads. How can you ensure all threads that started from main must end in order in which they started and main should end in fast? 2.5
   c) What are Java packages? What is the significance of packages. 3

8. a) Explain why design patterns in Java is necessary? 2
   b) Briefly describe creational design patterns and factory design pattern. 3
   c) Describe bridge design pattern with an example. 2
   d) Mention which pattern is useful when one has to pass data with multiple attributes in one shot from client to server? 1.75

University of Rajshahi

Department of Computer Science and Engineering

B.Sc. Engineering Part-I Even Semester Examination 2020

Course Code: CSE1221 Course Title: Object Oriented Programming

Time: 3 Hours **Full Marks: 52.5**

[N.B. Answer SIX questions taking at least THREE from each Section.]

Section- A

1.(a) Which of the following are legal identifiers in Java? 1.75

```
(i)g (ii)void (iii)101dalmatians (iv)Hello, World (v)<greeting>
```

(b) What does this sequence of statements print? 1.50

```
Rectangle box = new Rectangle(5, 10, 20, 30);
System.out.println("Before: " + box.getWidth());
box.translate(25, 40);
System.out.println("After: " + box.getWidth());
```

(c) What is encapsulation? Why is it useful? 2.00

(d) Consider the following implementation of a class Square : 2.00

```
public class Square{
private int sideLength;
private int area; // Not a good idea
     public Square(int length){
          sideLength = length;
     }
     public int getArea(){
          area = sideLength * sideLength;
          return area;
     }
}
```

Why is it not a good idea to introduce an instance variable for the area? Rewrite the class so that area is a local variable.

(e) Write the difference between equals() method and equality operator (==) in Java. 1.50

2.(a) What is vector? How is it different from an array? 2.75

(b) How does **String** class differ from the **StringBuffer** class? Explain with example. 3

(c) Write a method called 3

```
delete (String str, int m)
```

that returns the input string with the mth element removed.. 2.00

3.(a) What do these loops print by given the variables

```
String stars = "******";
String stripes = "=====";

(i) int i = 0;
while (i < 5){
     System.out.print(stars.substring(0, i));
     System.out.println(stripes.substring(i, 5));
     i++;
}
```

5.(a) Suppose the c
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```
(ii) int i = 0;
while (i < 10){
     if (i % 2 == 0)
          System.out.println(stars);
     else
          System.out.println(stripes);
}
```

1.75

(b) The nested loops

```
for (int i = 1; i <= height; i++) {
     for (int j = 1; j <= width; j++) {
          System.out.print("*");
     }
     System.out.println();
}
```

Display a rectangle for a given width and height, such as (width=4, height=3)

```
****
****
****
```

Write a single for loop that displays the same rectangle.

(c) Consider the following loop for collecting all elements that match a condition; in this case, that the element is larger than 100. 2.00

```
ArrayList<Double> matches = new ArrayList<Double>();
for (double element : values)
     if (element > 10)
          matches.add(element);
```

Trace the flow of the loop, where values contain the elements 10 9 10 12 8. Show two columns, for element and matches.

(d) How do you perform the following tasks with array lists in Java? 3.00

(i) Test that two array lists contain the same elements in the same order.
(ii) Copy one array list to another.
(iii) Fill an array list with zeros, over writing all elements in it.

4.(a) Why is it a good idea to minimize dependencies between classes? 1.75

(b) You need to write a program for DNA (Contain A, T, C & G) sequence analysis that checks whether a substring of one string is contained in another string. What simpler problem can you solve first? 2.00

(c) Give a rule of thumb for how to find classes and methods when designing a program. 2.00

(d) After discovering a method, why is it important to identify the object that is responsible for carrying out the action? 1.00

(e) Every BMW is a vehicle. Should a class BMW inherit from the class Vehicle ? BMW is a vehicle manufacturer. Does this mean that the class BMW should inherit from the class VehicleManufacturer? 2.00

Section-B

5.(a) Suppose the class Employee is declared as follows: 3.00

```
public class Employee{
      private String name;
      private double baseSalary;

      public void setName(String newName) { . . . }
      public void setBaseSalary(double newSalary) { . . . }
      public String getName() { . . . }
      public double getSalary() { . . . }
}
```

Declare a class `Manager` that inherits from the class `Employee` and adds an instance variable `bonus` for storing a salary bonus. Omit constructors and methods.

(b) Will the following code fragment compile? Will it run? If not, what error is reported? 1.75

```
Object obj = "Hello";
System.out.println(obj.length());
```

(c) What does this code fragment print? Why is this an example of polymorphism? 2.00

```
Measurable[] data =
{ new BankAccount(10000), new Country("Belgium", 30510) };
System.out.println(average(data));
```

Assume that the class `BankAccount`, `Country` and the interface `Measurable` has a proper definition.

(d) Suppose the class Sandwich implements the Edible interface, and you are given the variable declarations 2.00

```
Sandwich sub = new Sandwich();
Rectangle cerealBox = new Rectangle(5, 10, 20, 30);
Edible e = null;
```

Which of the following assignment statements are legal?

```
 i.   e = sub;
 ii.  sub = e;
 iii. sub = (Sandwich) e;
 iv.  sub = (Sandwich) cerealBox;
 v.   e = cerealBox;
 vi.  e = (Edible) cerealBox;
 vii. e = (Rectangle) cerealBox;
 viii. e = (Rectangle) null;
```

6.(a) What is wrong with the following code, and how can you fix it? 3.00

```
1. public static void writeAll(String[] lines, String
   filename){
2.       PrintWriter out = new PrintWriter(filename);
3.       for (String line : lines){
4.            out.println(line.toUpperCase());
5.       }
6.       out.close();
7. }
```

(b) The following code tries to close the writer without using a try-with-resources statement:

```
PrintWriter out = new PrintWriter(filename);
try{
      //Write output.
      out.close();
}catch (IOException exception){
      out.close();    }
```

What is the disadvantage of this approach?

(c) Suppose the input contains the characters 995.0 Fred. What are the values of number and input after this code fragment? 1.75

```
Scanner in = new Scanner(System.in);
int number = 0;
if (in.hasNextInt()) {
      number = in.nextInt();
}
String input = in.next();
```

(d) How do you move the file pointer to the first byte of a file? To the last byte? To the exact middle of the file? 2.00

7.(a) Show two different ways of converting **String** object into **Integer** in Java. 1

(b) An n-digit number that is the sum of the n-th powers of its digits is called an Armstrong Number. For example, 153 and 370 are Armstrong numbers, since $153 = 1^3 + 5^3 + 3^3$ and $370 = 3^3 + 7^3 + 0^3$. But 15 and 37 are not Armstrong numbers, since $15 \neq 1^2 + 5^2$ and $37 \neq 3^2 + 7^2$. First few Armstrong numbers are 1, 2, 3, 4, 5, 6, 7, 8, 9, 153, 370, 371, 407, 1634 and so on. Write a Java program to find whether the given number is Armstrong or not. 4.75

(c) Why is an Interface used in Java? How is an Interface declared and used? 3

8.(a) Consider the following runnable. 3.00

```
public class MyRunnable implements Runnable{
      public void run(){
            try{
                  System.out.println(1);
                  Thread.sleep(500);
                  System.out.println(2);
            }catch (InterruptedException exception){
                  System.out.println(3);
            }
            System.out.println(4);
      }
}
```

Suppose a thread with this runnable is started and immediately interrupted:

```
Thread t = new Thread(new MyRunnable());
t.start();
t.interrupt();
```

What output is produced?

(b) What are the two ways of implementing thread in Java? Provide necessary coding examples. 4.00

(c) How do threads communicate with each other? 1.75

University of Rajshahi
Department of Computer Science and Engineering
B. Sc. (Engg.) Part-1, Even Semester Examination 2019
Course: CSE-1221 (Object Oriented Programming)
Marks: 52.5 Duration: 3 hours

(Answer 3 questions from each part)

Part –A

1. a) What are the similarities and differences between Java and C++? 2.5

   b) What is the role of Java Virtual machine? 2.5

   c) In the following code snippet, explain what each of the keywords is used for. 3.75

```
public static void main (String []args ) throws IOException {
      // code inside main method
}
```

2. a) In the following Java code identify as many of the mistakes as you can. Explain whether each mistake you identify will cause compilation error, runtime error or logical error. 7

```
public Class mycode.java
{
    void static public fun main(String [args])
    begin
        Leaf tr = null;
        for (i=1; i>10; ++i)
            tr = new Node(i, tr)
            tr.print();
    end;
}

class Leaf {
    integer value;

    Leaf(int value){   this = value; }

    public void print(){   System.out.println(value);   }
}

class Node extends Leaf {
    Leaf left, right;

    Node(leaf l, Leaf r) {   left = l, right = r;   }

    void print(){
        left.print();
        System.out.println("val=" @ value);
        right.print();
    }
}
```

   b) Distinguish between overloading and overriding with examples in Java code.. 1.75

3. a) What are the differences and similarities between abstract class and Interface? Mention a situation when the use of one of them is preferable over the other. 3.5

   b) See the program below. Identify and explain the problems in the code. After fixing the problem, what will be the output? 3

```
class Base {
    final public void Print() {
        System.out.println("Base");
    }
}

class Derived extends Base {
    public void Print() {
        System.out.println("Derived");
    }
}

class Main{
    public static void DoPrint( Base o ) {
        o.Print();
    }
    public static void main(String[] args) {
        Base x = new Base();
        Base y = new Derived();
        Derived z = new Derived();
        DoPrint(x);
```

c) What are the purposes of **super** keyword in java?

4. a) What do you understand by object-oriented design pattern? Why should we use them? 2.75

b) Suppose that your program has a class that you want to have only one instance created and this instance must be reused during the lifetime of your program. Which design pattern will you use to implement that. Explain with example. 3

c) Briefly discuss the idea of Observer pattern with example. 3

Part –B

5 a) Explain exception handling mechanism in Java. 3

b) When is **throws** keyword is used in java? 1.5

c) Explain the hierarchy of exceptions in java? 3

d) What are checked and unchecked exceptions in java? 1.25

6. a) What is thread? Describe the life cycle of a thread. 4

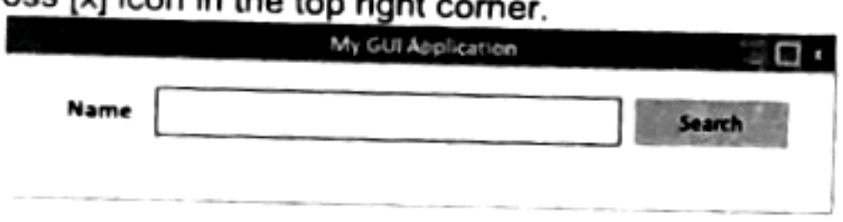
b) How do you ensure that n threads can access n resources without deadlock? 2

c) What is "Instance Variable Hiding" problem? How can this problem be solved? 2.75

7. a) What are the functions of Event object, event source and event listener in delegate event model? Explain with diagram. 3

b) what are the advantages of swing over AWT? 2

c) Write an AWT based application that uses **FlowLayout** manager to layout the following UI that can be closed by clicking cross [x] icon in the top right corner. 3.75



8. a) In a large project it is possible that two programmers, working on different parts of the code, happen to select the same name for a class that they define. If this accident leads to some third programmer accessing the wrong one of these two classes, the results could be unsatisfactory. Explain the steps that the designers of the Java language have taken to reduce the probability of such incident. 2

b) Explain what problem can arise while using the following generic class. How to fix it? 3

```
class Stats<T> {
  T[] nums;

  Stats(T[] o) {
    nums = o;
  }

  double sum() {
    double x = 0.0;
    for(int i=0; i < nums.length; i++)
      x += nums[i].doubleValue();

    return x;
  }
}
```

c) Write a java program that reads the content of a file and writes the content back to another file. 3.75

this
2.75
3.

University of Rajshahi
Department of Computer Science and Engineering

B. Sc. (Engg.) Part-1, Even Semester Examination 2018

Course: CSE-1221 (Object Oriented Programming)

Marks: 52.5 **Duration: 3 hours**

(Answer 3 questions from each part)



Part –A

1. a) Why java is called "the language of the Internet"? Explain. 2

   b) Compare Java with C programming language. 2

   c) Consider the following java program: 4.75

```
public interface IRunable {
    public void run();
}

Class JavaApp extends ClassA, ClassB  implements IRunnable
{
      void main(String *args) {
           System.in.println("This is a Java program.");
      }
}
```

   Point out the mistakes. Also, correct the program so that it can be compiled.

2. a) What is singleton class? 1.5

   b) What is Final keyword in Java? Give an example. 2.25

   c) Write a program in Java which take 10 numbers, store it in a combo box and display the largest one. 3

   d) What happens if an exception is not caught? 2

3. a) You are writing a program which can draw various 2D shapes like circle, rectangle, triangle and 3D shapes like cube and spheres of with different colors and calculate the area and volume of the shapes. 4.5

   With class diagram explain how you will achieve the task in an object-oriented manner?

   b) Briefly describe the **Object** class in java. 2.25

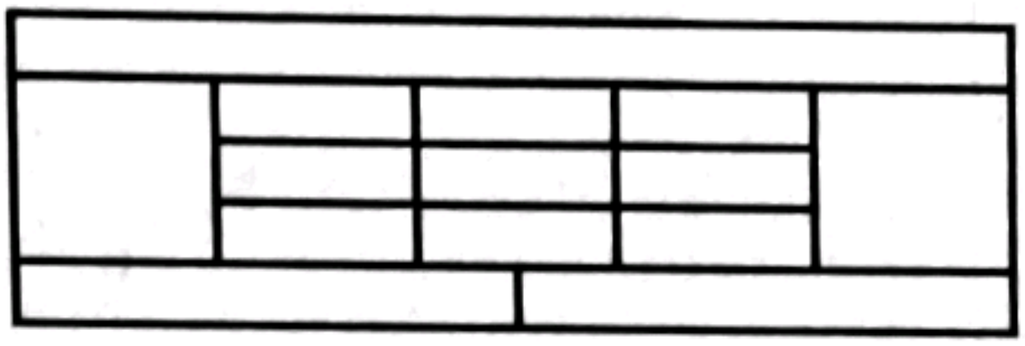
   c) What are the conditions for a class to become abstract? Can you declare an object of abstract class? 2

4.	a)	Discuss how **try-catch-finally** block works in java.	3
	b)	How can you define your own exception type in java?	2.5
	c)	What are the difference between **InputStream** class and **Reader** class?	1
	d)	Write a simple java program to copy content of a text file into another file.	2.25

Part –B

5	a)	What is thread? Discuss java thread model.	3.25
	b)	Write a program to demonstrate the creation and execution of threads. How can you ensure all threads that started from main must end in order in which they started and main should end in last?	2.5
	c)	What is "Instance Variable Hiding" problem? How this problem can be resolved?	3

6.	a)	With suitable block diagram explain Delegation Event Model in java.	3
	b)	Why is swing preferable for GUI programming over AWT?	2
	c)	What are the purpose of event Adapter classes?	1
	d)	Describe shortly how to achieve the following GUI layout. (Which layouts are used, how they are nested?)	2.75



7.	a)	What is Dynamic binding? Show with an example how dynamic binding works.	2
	b)	What are Java packages? What is the significance of packages?	3
	c)	Design a Java package for numbers. Develop two different classes that belongs to two package, one for to check whether the given number is palindrome or not and the other to check whether the given number is odd or even and access these package using one main file	3.75

8.	a)	What do you mean by "design pattern" in OOP? How are they useful?	2
	b)	With appropriate example and class diagram explain the Abstract Factory Design Pattern.	5
	c)	Which design pattern will you use to make sure only one instance of an object is created and reused during the execution cycle of a java program? Explain.	1.75

University of Rajshahi

Department of Computer Science and Engineering

B. Sc. Engg. Part1 Even Semester, Examination-2018 (Session 2016-17)

Course: CSE-1221 (Object Oriented Programming with C++)

Full Marks: 52.5 **Time: 3Hours**

[Answer any <u>six</u> questions taking <u>three</u> from each Section]

SECTION A

1.	a)	Briefly describe the key features in object oriented programming language?	3.75
	b)	Shortly describe the purpose of C++ namespace.	3
	c)	Define C++ class and objects.	2
2.	a)	How can you define the access rights for the members of a class? Write a simple C++ program using various access specifiers.	3.75
	b)	What are the purposes of constructor and destructor functions?	3
	c)	What is copy constructor in C++?	2
3.	a)	How can you give permission to a nonmember function access to the private members of a class?	4.75
	b)	Write down the usefulness of friend function.	2
	c)	How can you open and close a file in C++ program by using different types of streams?	2
4.	a)	Write a fragment of program that overload member operator function (+).	3
	b)	How can you create prefix and postfix forms of the increment and decrement operators?	3
	c)	Friend operator functions add flexibility. Discuss with example.	2.75

SECTION B

5. a) Define base class and derived class. Do base class and derived class have any specific purpose in C++ programming language to achieve any key feature in Object Oriented Programming? Explain with C++ code. 5.75

b) Are the protected members of a base class accessible by a derived class when the base class is inherited as public? 3

6. a) Write the output of the following C++ program. 4

```
#include <iostream>
using namespace std;
class base {
public:
        base() { cout << "Constructing base\n"; }
        ~base() { cout << "Destructing base \n; }
};
class derived: public base {
public:
        derived() {cout << "Constructing derived\n"; }
        ~derived() { cout << "Destructing derived\n"; }
};
int main() {
        derived ob;
        return 0; }
```

b) What is the purpose of using virtual base class? 2.75

c) What is exception handling? 2

7. a) How can we create initialized and uninitialized array of objects by overloading constructor function? Give example with C++ code. 4.75

b) Explain the use of pointer in C++. What do you mean by 'this' pointer? 4

8. a) What is generic function? Write the general form of a template function definition. 2

b) Can you mix standard parameters with generic type parameters in a template function? Explain. 3

c) Write the use of width(), precision() and fill() member functions with example. 3.75

University of Rajshahi
Department of Computer Science and Engineering
B.Sc.(Engg.), Part-1, Even Semester Examination-2017
Course: CSE-1221 (Object Oriented Programming with C++)
Total marks: 52.50 **Time: 3 hours**

[Answer **three** questions from each section]

Section-A

1. a) What is object oriented programming? Why is it effective over structured programming? Explain. 3
   b) Define the term object and class. How does class accomplish data hiding? 3
   c) Is it possible to pass an object to a function? If possible, how? 2.75

2. a) Can you create an initialized array of objects? Explain. 1.75
   b) What is reference parameter? Write a fragment of program to illustrate passing reference to object. 4
   c) How can you initialize allocated memory to some known value? Give example. 3

3. a) 'Constructor overloading can be used to gain flexibility' – explain. 2.75
   b) Explain the use of pointer in C++. What do you mean by '**this**' pointer? 4
   c) What are the differences between pointers to constants and constant pointers? 2

4. a) Define function overloading. What will be the output of the following code? 3.75

```
#include <iostream>
using namespace std;
int count = 0;
class A{
   public:
   A(){}
   void func(){cout<<"Fucntion calling A"<<endl;}
};
class B:public A{
   public:
   B(){}
   void func() {cout<<"Function calling B"<<endl;}
};
int main()
{
   A *a;
   B b;
   a=&b;
   a->func();
   return 0;
}
```

   b) Write a C++ program to define three overloaded functions to swap two integers, swap two floats and swap two doubles. 3
   c) Differentiate between function overloading and function templates. 2

Section-B

5. a) Class 'Base' has one public member function abc() and one protected member function xyz(). What will be happened if classes 'Derived1', 'Derived2', 'Derived3' inherit 'Base' class as public, private, and protected respectively?
   b) What is virtual function?
   c) 'Virtual functions are hierarchical' – explain it with an example.

6. a) Briefly discuss generic classes with example. 3
   b) What will be the output of the following code? 2.75

```
#include <iostream>
using namespace std;
int Created=0, Destroyed=0;
class Counter{
  public:
  Counter(){ Created++;}
  ~Counter(){ Destroyed++;}
};
void func(){
   Counter obj1, obj2, obj3;
cout<<Created<<" "<<Destroyed<<endl;
}
int main(void){
   Counter o1, o2;
cout<<Created<<" "<<Destroyed<<endl;
cout<<Created<<" "<<Destroyed<<endl;
func();
cout<<Created<<" "<<Destroyed<<endl;
   return 0;
}
```

   c) Define operator overloading with example. What are the restrictions applied to operator overloading? 3

7. a) What will be the output of the following code? Draw recursive tree for N=13. 2.75

```
#include <iostream>
using namespace std;
int count=0;
class Recursive{
   public:
intfunc(int n){
      if(n==0) return 0;
      if(n%2==0) return 1+func(n/2);
      else return 1+func(n-1);
   }};
int main(){
   Recursive rec;
cout<<rec.func(13) <<endl;
   return 0;
}
```

   b) What is exception handling? Explain types of exception handling. 3
   c) What is difference between exceptions and error? 1
   d) What is Standard Template Library? 2

8. a) Briefly discuss different file open mode. 2.75
   b) What are purposes of these following functions: getline(), EOF(), ignore(), peek(), putback() and flush(). 3
   c) How can you access a file randomly? Give example. 3

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University of Rajshahi
Department of Computer Science and Engineering
B. Sc. Engg. Part1 Even Semester, Examination-2016
Course: CSE-1221 (Object Oriented Programming with C++)
Full Marks: 52.5 **Time: 3 Hours**
Answer three questions from each part

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Part A

1.	(a) Write some features of OOP.	2
	(b) Differentiate between structure and class with an example.	1.75
	(c) What is local class? What are the restrictions of local classes? Discuss with example.	3
	(d) Is it possible to return an object from a function? If possible, how?	2
2.	a) Is it possible to have arrays of objects? If possible, how?	2
	b) What is 'this pointer'? Explain.	2
	c) Define inline function. Discuss about the restrictions with inline function.	3
	d) How can you allocate objects dynamically?	1.75
3.	a) What is constructor overloading? What are the reasons to overload a constructor?	2.75
	b) Why overloading sometimes causes ambiguity? Describe with example.	3
	c) What is default argument? What are the advantages of default arguments? Give example.	3
4.	a) Explain different types of inheritance with block diagram and examples.	3.5
	b) What is the ambiguity that arises in multiple inheritance? How it can be overcome? Explain with example.	2.5
	c) Discuss the implications of deriving a class from an existing class by the 'public' and 'protected' access specifiers with examples.	2.75

Part B

5.	(a) Why is constructor functions executed in order of derivation?	1
	(b) How do you pass arguments to a constructor in a base class? Discuss with example.	4
	(c) Briefly discuss virtual base class with an example.	3.75
6.	(a) 'The virtual attribute is inherited'- explain it with an example.	2.75
	(b) What is a pure virtual function? Discuss it with an example.	3
	(c) Differentiate between early binding and late binding.	3
7.	(a) How can you define more than one generic data type in the template function? Explain with an example.	2
	(b) Can you mix standard parameters with generic type parameters in a template function? Explain.	2
	(c) Is it possible to catch a class type? Explain.	2.75
	(d) How can you restrict the type of exceptions that a function can throw outside of itself?	2
8.	(a) List and explain in brief the various functions required for random access file operations.	3.5
	(b) How do 'endl' and 'setw' manipulators work? Illustrate with example.	2.5
	(c) What is the difference between opening a file with constructor function and opening a file with 'open()' function?	2.75

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. (Engg.) Part-I Even Semester Examination-2015
Course: CSE1221 (Object Oriented Programming with C++)

Marks: 52.5 **[Answer any six (06) questions taking three (03) from each part]** **Time: 3 Hours**

Part-A

1. a) What is object-oriented programming? How are data and functions organized in an object-oriented program? 3
   b) What kind of things can become an object in OOP? 1.75
   c) Distinguish between the following terms: 3
      i) Inheritance and polymorphism
      ii) Objects and classes
   d) What is meant by data binding? 1

2. a) How we can run our C++ program without header files? 1.75
   b) Describe the major parts of a C++ program. 3
   c) How can you create an initialized array of objects? 2
   d) What is new and delete? What are their advantages? 2

3. a) Briefly discuss copy constructor with an example. 4
   b) Write some ambiguities of function overloading. 2.75
   c) What is a constructor? What is wrong with the constructor shown in the following fragment? 2
   class sample { double a, b, c;
   public: double sample(); };

4. a) How does binary operator operate? Give an example. 3
   b) How can you create prefix and postfix forms of the increment and decrement operators? 2.75
   c) Write a fragment of program that overloads friend operator function (-). 3

Part-B

5. a) What are the different forms of inheritance? Give an example of each. 4.75
   b) In what order are the class constructors called when a derived class object is created? 2
   c) Consider the following code: 2
   class B { };
   class D1 : public B { };
   class D2 : public B { };
   class DD : public D1, public D2 { };
   How can you prevent the creation of two copies of base class B in a DD object?

6. a) How is polymorphism achieved at (a) compile time and (b) run time? 2
   b) When do we make a virtual function "pure"? What are the implications of making a function a pure virtual function? 3
   c) How could we specify the types of objects a function can throw? 2
   d) When should a function throw an exception? 1.75

7. a) What is exception handling? Write the general form of exception handling. 3
   b) Is it possible to write a try block inside a function? Explain. 2
   c) Write the use of width(), precision() and fill() member functions with example. 3.75

8. a) How can you read from and write to a text file? Discuss. 3
   b) Briefly discuss getline() function with example. 2
   c) Shortly discuss random access from a file with example. 3.75

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University of Rajshahi

Department Of Computer Science and Engineering

B.Sc.(Engg.) Part-I, Even Semester Examination-2014

Course: CSE1221 (Object Oriented Programming with C++)

Full Mark: 52.5 Duration: 3 Hours

[N.B. Answer any six questions taking three from each group]

Part-A

1. a) What is object oriented programming? Why is it effective over structured programming? Explain. 2.75
   b) Briefly discuss the properties of OOP language with example. 6

2. a) Write the general form of a class. Differentiate between class and object. 3
   b) What is friend function? Why is it used? Describe with an example. 3
   c) What is in-line function? What are the restrictions to in-line functions? 2.75

3. a) Can an array of objects be initialized? Explain. 3
   b) In C++ there are two ways to achieve call by reference. What are they? Discuss with example. 2
   c) How can you allocate and free memory to an object? Explain with example. 3.75

4. a) What are constructor and destructor function? Can these functions have input parameters and return type? 3
   b) There are some errors in the following program code. Find the errors and rewrite the corrected program and print the output of the program 3

```
class A
{ int x;
public:
int A();
void set(int b);
int get();
}

void set(int b)
{ x=b;}
int A::get()
{ return x;}

main()
{
A ob1,ob2;
ob1.x=10;
ob2.set(11);
cout<<"The value of
x="<< ob1.get()
<<"and" <<ob2.get(); }
```

   c) What are the advantages of using reference over pointer? What are the restrictions of using reference? 2.75

B
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Time: 3 H
Answer

Part-B

5.	a)	What are the differences between member operator function and friend operator function?	2
	b)	In case of member operator function, how does binary operator operate? Give an example.	2.75
	c)	Write a fragment of code that make use of the shorthand operators like += and -=.	2
	d)	Explain with example, how does friend operator functions add flexibility.	2
6.	a)	What is multi-level inheritance? Describe a scenario in which multi-level inheritance can cause ambiguity.	2.5
	b)	How the ambiguity mention above can be solved?	1.25
	c)	Write a short program that will use C++ I/O library to write ***n*** random numbers to a file.	3
	d)	What is an abstract class? Can we create instances of an abstract class?	2
7.	a)	Briefly discuss generic function. Write the general form of a template definition.	2.75
	b)	How can you overload a generic function explicitly? Explain with example.	3
	c)	How can you mix standard parameters with generic type parameters in a template function? Explain.	3
8.	a)	What is exception handling?	1
	b)	Briefly discuss try, catch and throw.	3
	c)	What is the mechanism to catch all the exceptions?	1.75
	d)	How can you rethrow an exception? Explain with example.	3

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engg. Part-1 Even Semester, Examination-2013
Course: CSE-1221 (Object Oriented Programming with C++)
Time: 3 Hours Full Marks: 52.5

Answer **six** questions taking any **three** questions from each section

SECTION: A

1	(a) What is meant by Object Oriented Programming (OOP)? Write the features of OOP.	2+1
	(b) How data hiding is accomplished in C++? Explain.	3
	(c) How are data and functions organized in OOP? Explain.	2.75
2	(a) Differentiate between C and C++. When to use C++?	2+1
	(b) Write the general program structure of C++ and explain each section.	3
	(c) Explain about different phases for executing a C++ program?	2.75
3	(a) Write different data types used in C++. What is data abstraction?	3
	(b) Discuss class and Object with example.	3
	(c) Explain the accessing mechanism of data members and member functions in case of (i) inside main () function and (ii) inside a member function of the same class.	2
	(d) When do you declare a member of a class static?	0.75
4	(a) Can an array of objects be initialized? Explain.	2
	(b) What is "this"? Explain with an example.	2
	(c) What is new and delete? What are their advantages?	2.75
	(d) What is reference? Describe with example.	2

SECTION: B

5	(a) Briefly discuss inheritance with an example.	4
	(b) Why is destructor function executed in reverse order of derivation?	1
	(c) What are the two ways in which a derived class can inherit more than one base class?	3.75
6	(a) Explain different types of polymorphism in C++.	4
	(b) Create an abstract base class shape with two members base and height, a member function for initializing and a pure virtual function to compute area(). Derive two specific classes Triangle and Rectangle which override the function area (). Use these classes in a main function and display the area of a Triangle and a Rectangle.	3
	(c) Distinguish between static binding and dynamic binding.	1.75
7	(a) What is stream? Name the streams generally used file I/O.	2
	(b) Explain various file modes used in C++.	3
	(c) Explain the various file stream classes needed for file manipulations in C++.	1.75
	(d) Explain the functions seekg, seekp, tellg, tellp used for setting pointers during file operation.	2
8	(a) What is an exception? Why do we need to handle exception?	1+1
	(b) What are the keywords used in C++ for executing handling? Describe their usage with suitable example.	1+2
	(c) How can you create a function template in C++?	3
	(d) Give general syntax of a generic class.	0.75

University of Rajshahi
Department of Computer Science and Engineering
B.Sc.Engg.(CSE) 1st Year Even Semester 2012
Course: CSE1221 (Object Oriented Programming with C++)
Full Marks: 52.5 Time: 04 hours
[N. B. Answer any SIX questions taking at least THREE each part.]

Part A

1. a) **What is a class? What is an object?** 1.75

b) **What is a nested class? Why can it be useful?** 1

c) **Find the error(s) in each of the following and explain how to correct it:** 3

i. `void ~Time( int );`

ii. The following is a partial definition of class Time:

```
class Time{
public:
// function prototypes
Private:
int hour=0;
int minute=0;
int second=0;
```

iii. Assume the following prototype is declared in class Employee:

```
int Employee( const char *, const char *);
```

d) **Find the errors in the following class and explain how to correct them:** 3

```
Class Example{
public:
Example(int y=10)
    : data(y)
 {
    //empty body
 }
 int getIncrementedData() const
 {
    Return ++data;
 }
 static int getCount()
 {
    cout<<"Data is "<<data<<endl;
    return count;
 }
 private:
    int data;
    static int count;
}; // end class Example
```

a) **What is inline function?** 1.5

b) **How do you tell the compiler to make a member function inline?** 1.5

c) Why might the following function not be in-lined by your compiler? 1.75

```
void f1()
{
    int i;
    for(i=0;i<10;i++) cout<<i;
}
```

d) Given the following partial class, add the necessary constructor functions so that both 4 declarations within **main()** are valid.

```
class samp{
    int a;
public:
    //add constructor functions
    int get_a(){return a;}
};
int main(){
    samp ob(88); // init ob's a to 88
    samp obarray[10]; // noninitialized 10-element array
    //...
}
```

3. a) **Define constructor and destructor?** 1.75

b) **What is copy constructor? When copy constructors called?** 1

c) **What is the difference between the effects of the following two declarations:** 1

```
Ratio y(x);
Ratio y=x;
```

d) **What is the difference between the effects of the following two lines:** 1

```
Ratio y=x;
Ratio y; y=x;
```

e) To illustrate exactly when an object is constructed and destructed when returned from a function, create a class called **who**. Have **who**'s constructor take one character argument that will be used to identify an object. Have the constructor display a message similar to this when constructing an object: 4

```
Constructing who#x
```

Where x is the identifying character associated with each object. When an object is destroyed, have a message similar to this displayed:

```
Destroying who#x
```

Where, again, x is the identifying character. Finally, create a function called **make_who()** that returns a **who** object. Give each object a unique name. Note that output displayed by the program.

4. a) **What do you mean by operator overloading? What are the restrictions applied to operator overloading?** 2.75

b) What are the advantages of overloading an operator using friend function? 3

c) Explain why protected access specifier is used while inheriting a base class? 2

d) In derived class constructors and distractors are called in which order? 1

Part B

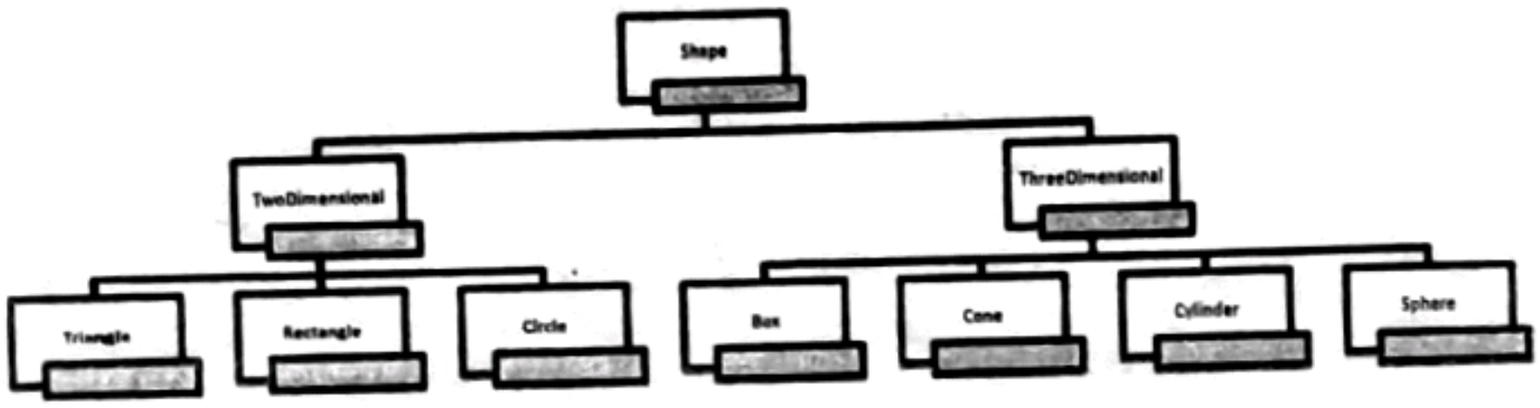
a) What is a friend? Do friends violate encapsulation? 1.75
b) What are some advantages/disadvantages of using friend functions? 1.5
c) Why should I use <iostream> instead of the traditional <cstdio>? 2
d) What is "this" pointer? 1.5
e) Write the output for the following lines 2

```
std::cout << i << '\n';
std::cout << "i" << '\n';
std::cout << i / j << '\n';
std::cout << "i=" << i;
```

2. a) What is multiple inheritance (virtual inheritance)? What are its advantages and disadvantages? 1.5
b) What is a "virtual member function"? 1
c) What is pure virtual function? What is abstract class? 1.5
d) What is wrong with the following definitions: 1.75

```
class X
{protected:
      Int a;
};
class Y: public X
{public:
      Void set(X x, int c) { x.a=c;}
}
```

e) Implement the following class hierarchy: 3



3. a) Distinguish between runtime and compile time polymorphism? How runtime polymorphism is achieved? 3.75
b) What is the difference between virtual function and overloaded function? 2
c) What is a pure virtual function? How a pure virtual function is declared? 1
d) Define early binding and late binding with example. 2

a) What's the idea behind templates? What's the syntax / semantics for a "class template"? 1

b) Define a Swap function template for swapping two objects of the same type 1

c) What are some ways try / catch / throw can improve software quality? 1.75

d) How should I handle resources if my constructors may throw exceptions? 1

e) Consider the following function which is used for receiving a packet in a network system: 4

```
void ReceivePacket (Packet *pack, Connection *c)
{
        switch (pack->Type()) {
             case controlPack: //...
                  break;
             case dataPack: //...
                  break;
             case diagnosePack: //...
                  break;
             default: //...
        }
}
```

Suppose we wish to check for the following errors in *ReceivePacket:*

- That connection c is active. `Connection::Active()` will return true if this is the case.
- That no errors have occurred in the transmission of the packet. `Packet::Valid()` will return true if this is the case.
- That the packet type is known (the `default` case is exercised otherwise).

Define suitable exceptions for the above and modify `ReceivePacket` so that it throws an appropriate exception when any of the above cases is not satisfied.

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Department of Computer Science and Engineering

B.Sc. Engg.(CSE) 1st Year 2nd Semester 2011

Course: CSE 1221 (Object Oriented Programming)

Time: 4 Hrs. **Full Marks: 52.5**

[N.B. Answer SIX questions taking at least THREE from each part.]

Part-A

1. a) Define Object Oriented Programming (OOP). What are features of OOP? 3.75
   b) What are differences between OOP and Procedural Programming? 3
   c) What are the differences between 'class' and 'structure'? 2

2. a) What are 'class' and 'object'? 2
   b) How are memory allocated for different type of members of a class? 2
   c) What are advantages of using 'static member variable' and 'static member function' of a class? 2.75
   d) Give an example of 'nested class member'. 2

3. a) What is operator overloading? Why is it necessary? 2
   b) Why overloading sometimes caused ambiguity? Which type of ambiguity may arise? 3
   c) What is the difference between early binding and late binding? 1.75
   d) What is default argument? 3.75

4. a) Is there any way to access private member of a class without taking help of own member function of that class? Explain your answer with an example. 2
   b) What are multiple, multilevel and hybrid inheritance? 3
   c) Explain how 'virtual inheritance' can solve the problem that is caused when any member of base class may be inherited in different ways to a 'high level derived class'. 2

Part-B

5. a) Can pointer to a base class be assigned to a pointer to derived class, or, vice versa? 3
   b) If both the base class and derived class have a member of same name, which member of the respective class can be and can't be, accessible by the above pointer assignment. 3.75
   c) Explain. with an example, what is the role of 'virtual function' for above situation. 2

6. a) What are 'unary' and 'binary' operator overloading? 3
   b) Why, if friend function is not used, then, left-side argument of binary operator can neither be built-in data type nor be an object of the class other than class having the respective overloaded operator function. 3.75
   c) Give an example of 'unary' operator overloading. 2.75

7. a) How is Java more secured than other languages? 2
   b) Why is Java known as platform-independent language? 4
   c) What types of naming convention should be followed when declaring package, variables, methods, and constant? 2

8. a) What is 'function template'? 2.75
   b) Give an example of 'function template'. 2
   c) Explain how exception is handled by using 'try', 'catch' and 'throw'? 2
   d) What are Java package and Java applets?